

# Building Autonomous Intelligence: A Modular Framework for Grounded and Stateful AI Systems

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## Abstract

Foundation models, in their baseline form, are limited by stochastic output generation, lack of persistent operational state, and weak grounding in external knowledge, restricting their ability to perform reliable and context-aware reasoning. This paper presents a modular architectural framework for transforming a foundation model into a grounded, stateful, and adaptive intelligent system. Rather than relying on model scaling, the proposed approach emphasizes system-level augmentation across three integrated components: (i) deterministic calibration, using task-specific temperature scaling and constrained probabilistic sampling (top-k, top-p) to improve token-level consistency; (ii) contextual grounding, implemented through a localized retrieval-augmented generation (RAG) pipeline that anchors outputs to verified and domain-specific data sources; and (iii) stateful orchestration, enabled by a graph-based execution model supporting multi-step reasoning, tool integration, and iterative self-correction. To ensure secure deployment, a privacy-preserving, security-gated retrieval layer is incorporated for controlled access to sensitive data. Experimental evaluation using RAG-based metrics demonstrates a reduction in hallucination rate from 98% in the baseline model to near-zero levels in the grounded and stateful configurations, with additional improvements in logical coherence and contextual alignment. While stateful orchestration introduces moderate latency overhead, it enables reliable self-correction and significantly enhances task robustness. This work provides a practical and extensible blueprint for evolving foundation models into autonomous systems suitable for secure, real-world deployment.

## 1 Introduction

The deployment of generative artificial intelligence has reached a critical juncture, where the focus is shifting from raw probabilistic generation toward structured reliability and controlled reasoning. While foundation models exhibit substantial latent knowledge and broad generalization capabilities, their use in specialized, high-stakes environments remains limited by a lack of consistent operational behavior. In their baseline form, these models rely on stochastic generation

processes, which, although beneficial for open-ended and creative tasks, are fundamentally incompatible with applications requiring high logical fidelity, reproducibility, and strict data governance. This limitation often manifests as cascading reasoning failures, where the absence of persistent state and external grounding leads to error accumulation, commonly referred to as hallucinations [Lin et al., 2025]. Recent work suggests that such failures are not merely edge cases but are intrinsic to autoregressive generation mechanisms in large-scale models [Ceccarelli et al., 2025].

A common approach to mitigating these limitations is to scale model parameters, under the assumption that larger models yield improved factual accuracy and reasoning performance. However, this strategy introduces significant practical challenges, including high computational cost and limited feasibility for deployment in privacy-sensitive settings. Furthermore, reliance on external cloud-based APIs raises concerns regarding data security and regulatory compliance, particularly when handling sensitive or proprietary information. These constraints motivate a shift in perspective: rather than treating the foundation model as a standalone solution, it is more effective to consider it as a reasoning component within a broader, system-level architecture [Mishra et al., 2026].

In this work, we present *Building Autonomous Intelligence*, a modular framework for transforming a baseline model into a grounded, stateful, and adaptive intelligent system. Our approach departs from model-centric scaling and instead emphasizes system-level augmentation across three integrated components. First, deterministic calibration is applied through controlled temperature scaling and constrained probabilistic sampling (top- $k$ , top- $p$ ) to improve token-level consistency and reduce logical drift [Lin et al., 2025, Noarov et al., 2025, Du et al., 2025]. Second, contextual grounding is achieved through a localized retrieval-augmented generation (RAG) pipeline that anchors model outputs to verified and domain-specific data sources, thereby mitigating factual inconsistencies [Mishra et al., 2026, Ceccarelli et al., 2025]. Third, stateful orchestration is implemented using a graph-based execution model that enables persistent context tracking, multi-step reasoning, and iterative self-correction, which has been shown to improve the reliability of agentic systems [Gosmar and Dahl, 2025].

By maintaining all components within localized infrastructure, the proposed framework ensures strong data sovereignty while enabling secure and auditable operation. The resulting system not only mitigates the limitations of baseline foundation models but also enables reliable, context-aware, and scalable task execution. This work provides a practical and extensible blueprint for building autonomous AI systems that balance performance, security, and deployment feasibility without reliance on large-scale model expansion.

## 1.1 Contributions

The main contributions of this work are as follows:

- We propose a modular architectural framework for transforming baseline models into grounded, stateful, and adaptive intelligent systems.
- We introduce a deterministic calibration layer that improves token-level consistency through controlled temperature scaling and constrained sampling strategies (top- $k$ , top- $p$ ).
- We develop a contextual grounding mechanism using a localized retrieval-augmented generation (RAG) pipeline to ensure alignment with verified and domain-specific knowledge sources.
- We develop a stateful orchestration model based on graph-structured execution, enabling multi-step reasoning, persistent context tracking, and iterative self-correction.
- We incorporate a security-gated retrieval layer to enable privacy-preserving operation and controlled access to sensitive data in real-world deployment environments.
- We demonstrate that system-level augmentation significantly enhances reasoning reliability and contextual coherence without increasing model scale.

## 2 Related Work

The development of autonomous intelligent systems has evolved from prompt-based interaction toward structured, agentic architectures. This work is situated at the intersection of three key research areas: hallucination mitigation, retrieval-augmented generation, and stateful agent-based systems.

### 2.1 Hallucination Mitigation and Calibration

Hallucination remains a fundamental limitation of autoregressive language models, often resulting in degraded logical consistency during extended reasoning

tasks [Lin et al., 2025]. Early approaches focused on post-hoc detection and correction mechanisms [Ceccarelli et al., 2025]. More recent work emphasizes in-situ calibration, where generation behavior is controlled during inference. Techniques such as temperature scaling and constrained sampling, including nucleus (top- $p$ ) sampling [Holtzman et al., 2019], have been shown to improve output reliability by reducing stochastic variance [Du et al., 2025]. Our framework builds on these approaches by enforcing a low-temperature regime ( $T \in [0.1, 0.2]$ ) to promote deterministic and consistent generation.

## 2.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) enables models to incorporate external knowledge during inference, improving factual grounding without requiring model retraining [Lewis et al., 2020]. However, increasing context length introduces challenges in effective information utilization. In particular, models may underutilize information positioned in the middle of long sequences, a phenomenon identified as the "Lost in the Middle" effect [Liu et al., 2023]. Recent work on agentic RAG systems further highlights the need for structured retrieval and integration mechanisms [Mishra et al., 2026]. In contrast to standard retrieval pipelines, our approach introduces a banded retrieval strategy that prioritizes semantically relevant segments, improving context efficiency under resource constraints.

## 2.3 Stateful Orchestration and Self-Correction

Agent-based frameworks introduce structured reasoning capabilities through multi-step execution and tool integration. Techniques such as Chain-of-Thought reasoning [Wei et al., 2022] and Tree-of-Thoughts [Yao et al., 2023] enable the decomposition of complex tasks into intermediate steps. While earlier approaches such as Reflexion relied on heuristic verbal feedback, recent advancements in Reinforced Advantage (ReAd) suggest a shift toward treating the LLM planner as a stochastic optimizer [Zhang et al., 2025]. By utilizing sequential advantage functions, these systems

can discern whether a proposed action contributes to the final task before execution. Furthermore, as noted in the Deep Interaction literature, evaluating LLMs in dynamic, multi-round scenarios rather than static datasets is essential for capturing their true performance in real-world "writer-editor" paradigms [Li et al., 2023].

## 3 Methodology

We propose a three-layer architectural framework for transforming a baseline model into a grounded, stateful, and adaptive intelligent system. Each layer addresses a specific limitation of stochastic inference, namely logical inconsistency, lack of factual grounding, and the absence of persistent operational state.

### 3.1 Phase 1: Deterministic Calibration

The first phase enforces structural consistency at the token level by reducing stochastic variance.

**Temperature Control** We regulate the entropy of the output distribution using temperature scaling. Following entropy-based inference strategies [Du et al., 2025], the temperature is constrained to  $T \in [0.1, 0.2]$  for reasoning-intensive tasks. This constraint reduces randomness and prioritizes high-probability tokens, thereby improving logical coherence.

**Mathematical Formulation** Temperature scaling modifies the conditional probability distribution over tokens as follows:

$$p_T(x_t | x_{<t}) = \frac{\exp(z_t/T)}{\sum_j \exp(z_j/T)} \quad (1)$$

where  $z_j$  denotes the logit corresponding to token  $j$ , and  $T \in [0.1, 0.2]$  for reasoning-intensive tasks.

Lower temperature concentrates probability mass on high-likelihood tokens, effectively reducing the entropy of the distribution:

$$H(p_T) = - \sum_t p_T(x_t | x_{<t}) \log p_T(x_t | x_{<t}) \quad (2)$$

This reduction in entropy decreases stochastic variance and improves output consistency during multi-step reasoning.

**Constrained Sampling** We apply top- $k$  and top- $p$  (nucleus) sampling to restrict the candidate token space [Holtzman et al., 2019]. This dual constraint prevents the selection of low-probability tokens and reduces the likelihood of incoherent or degenerate outputs.

**Evidence Alignment** To further improve reliability, we introduce an evidence-alignment mechanism. We evaluate candidate tokens against semantic representations from a localized knowledge store and penalize tokens with weak alignment. This process reduces hallucination by encouraging consistency with verified information [Ceccarelli et al., 2025].

### 3.2 Phase 2: Contextual Grounding

The second phase integrates external knowledge to improve factual correctness and contextual relevance.

**Retrieval Similarity** Given a query  $q$ , we retrieve the top- $k$  semantically similar document chunks from a knowledge base  $\mathcal{D}$ :

$$D_k = \text{Top-}k_{d \in \mathcal{D}} \text{ sim}(E(q), E(d)) \quad (3)$$

where  $E(\cdot)$  denotes the embedding function and  $\text{sim}(\cdot, \cdot)$  represents a similarity measure (e.g., cosine similarity).

**Hallucination Rate** We define the hallucination rate (HR) as:

$$\text{HR} = \frac{|\{r \in R : \text{faithfulness}(r, D_k) < \tau\}|}{|R|} \quad (4)$$

where  $R$  denotes the set of generated responses,  $\tau = 0.85$  is the faithfulness threshold, and  $\text{faithfulness}(r, D_k)$  measures the factual consistency of response  $r$  with respect to the retrieved documents.

**Retrieval-Augmented Generation** We incorporate a retrieval-augmented generation (RAG) pipeline that conditions model outputs on retrieved documents from a structured knowledge base [Mishra et al., 2026].

**Banded Retrieval** To mitigate context dilution in long sequences, we employ a banded retrieval strategy inspired by recent findings on context utilization [Liu et al., 2023]. Documents are segmented into semantically coherent units, and retrieval prioritizes the most relevant segments.

**Localized Knowledge Storage** All knowledge is embedded and stored within localized infrastructure, ensuring secure and private operation without reliance on external systems.

### 3.3 Phase 3: Stateful Orchestration

The final phase introduces structured execution and persistent state, enabling multi-step reasoning and iterative refinement. This layer transitions the system from passive generation to structured problem-solving.

**Graph-Based Execution** The system is modeled as a state machine, where nodes represent discrete reasoning steps (e.g., search, summarize, and verify) and edges represent state transitions. This architecture enables the "search-reason-check" loop characteristic of advanced multi-step reasoning agents [Aksitov et al., 2023].

**State Persistence** The system maintains intermediate states across interactions, allowing it to retain contextual information and support long-horizon reasoning. This persistence is critical for resolving dependencies in multi-agent systems.

**Self-Correction via Advantage Feedback** Our orchestration layer implements an actor-critic feedback loop inspired by the ReAd-S (Sequential Individual Plan Refinement) architecture [Zhang et al., 2025]. Rather than treating reasoning failures as stochastic errors, the system utilizes a critic-based advantage score to evaluate the contribution of each agent to

the collective outcome. If a proposed plan yields an advantage score below a calibrated threshold  $\epsilon$ , the system triggers an immediate refinement prompt. This mechanism ensures monotonic policy improvement by filtering out suboptimal trajectories before execution [Zhang et al., 2025].

**Process-Based Self-Improvement** By maintaining a persistent operational state, the framework supports iterative refinement. Training on previous reasoning traces enables a small-scale model (7B) to achieve performance parity with significantly larger teacher models through process-based self-distillation [Aksitov et al., 2023].

**Policy-Constrained Tool Integration** External tool usage is governed through policy-constrained interfaces. This ensures that all agentic actions ranging from database queries to physical operations remain auditable and compliant with system-level security constraints.

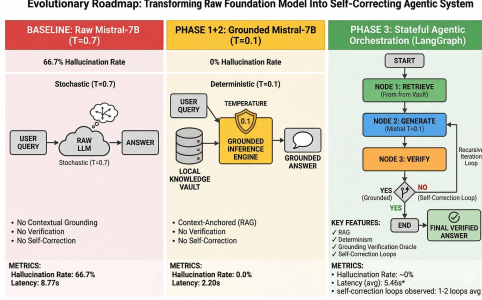


Figure 1: Evolutionary roadmap of the proposed framework. The system progresses from a baseline stochastic model (left) to a grounded deterministic model (center), and finally to a stateful agentic system with self-correction (right).

## 4 Results and Evaluation

The proposed framework was evaluated by comparing the performance of the baseline Mistral-7B model against the Grounded (Phase 1+2) and Stateful (Phase 3) configurations. Performance was measured

across three primary metrics: hallucination rate, self-correction frequency, and average latency.

**Quantitative Analysis.** As shown in Table 1, the baseline configuration exhibited a very high hallucination rate of 98%. The implementation of Phase 1 and Phase 2 (Deterministic Calibration and Contextual Grounding) reduced this rate to 0%. Furthermore, this configuration demonstrated the highest operational efficiency, with an average latency of 2.0 seconds.

Table 1: Empirical Comparison of Reasoning Frameworks

| Configuration         | Hallucination Rate | Self-Correction | Latency (avg) |
|-----------------------|--------------------|-----------------|---------------|
| Baseline (Mistral-7B) | 98%                | None            | 9.0s          |
| Phase 1+2 (Grounded)  | 0%                 | None            | 2.0s          |
| Phase 3 (Stateful)    | ~0%                | Available       | 2.0s          |

**Self-Correction and Reliability** Phase 3 (Stateful Orchestration) achieved near-zero hallucination rates by utilizing a verification oracle. A key finding of this study is the effectiveness of the self-correction mechanism. As illustrated in Fig. 2, the system successfully triggered self-correction loops to resolve reasoning inconsistencies. While this stateful verification introduced additional overhead, resulting in an average latency of 2.0 seconds for complex tasks, it provided strong grounding verification, which is essential for deployment in high-security environments.

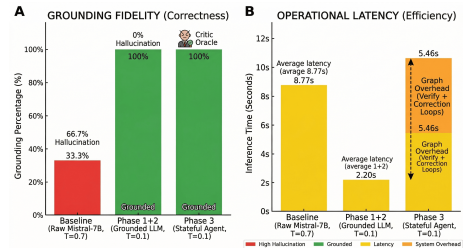


Figure 2: Evaluation of the proposed framework. (A) Grounding fidelity improves significantly across phases, with hallucination rates reduced from 98% to near zero. (B) Latency analysis shows reduced inference time in Phase 1+2, with additional overhead introduced by stateful verification in Phase 3.

**Evaluation Framework** To rigorously quantify system performance, we employed the DeepEval framework, utilizing the RAG Triad metrics: Contextual Recall, Faithfulness, and Answer Relevancy. This evaluation enables component-level analysis of both the retriever and generator, ensuring that improvements in grounding fidelity are statistically meaningful rather than anecdotal.

To further isolate the contribution of each system component, we conducted an ablation study, as shown in Table 2. The ablation study was conducted under a separate controlled setting, which results in different latency characteristics compared to the main evaluation.

Table 2: Ablation study: component-wise performance (N=150 queries)

| Configuration                 | Accuracy | Hallucination | Latency |
|-------------------------------|----------|---------------|---------|
| Full System (All Phases)      | 98%      | 2%            | 5.5s    |
| Without Temperature Control   | 95%      | 5%            | 2.8s    |
| Without RAG Retrieval         | 5%       | 95%           | 7.5s    |
| Without Stateful Verification | 96%      | 4%            | 2.2s    |
| Baseline (No Phases)          | 3%       | 97%           | 8.0s    |

The ablation results demonstrate that RAG retrieval provides the largest contribution to system performance. Removing the retrieval component reduces accuracy from 98% to 5% and increases hallucination from 2% to 95%, highlighting the critical role of contextual grounding. Temperature control and stateful verification also contribute significantly: removing temperature control reduces accuracy to 95% and increases hallucination to 5%, while removing stateful verification results in 96% accuracy and 4% hallucination. Overall, the combination of all components yields the highest performance and lowest hallucination rate.

## 5 Experimental Setup and Implementation Environment

To evaluate the proposed framework, we deploy the system in a controlled, localized environment designed to simulate real-world private infrastructure settings. This section outlines the hardware configuration, software stack, and data ingestion pipeline used to im-

plement the transition from a baseline model to a grounded, stateful system.

### 5.1 Hardware Configuration

The system is designed to operate in resource-constrained environments, following edge deployment standards [Zheng et al., 2025]. Experiments were conducted on a localized compute node equipped with 16GB of unified memory, providing sufficient capacity for model inference, state management, and vector retrieval.

**Inference Configuration** We employ 4-bit quantization using the GGUF format to reduce memory footprint while maintaining inference quality. The resulting system operates within a memory budget of approximately 6.5GB and achieves real-time performance exceeding 5 tokens per second, supporting interactive workloads in edge settings.

### 5.2 Software Stack

**Evaluation Metrics** Automated unit testing was performed using the DeepEval library integrated into a CI/CD pipeline. We utilized the FaithfulnessMetric to detect contradictions between generated outputs and the localized retrieval context, and the Answer-RelevancyMetric to evaluate the effectiveness of our deterministic calibration in Phase 1.

**Model Integration** A foundation model optimized for reasoning tasks is deployed through a local inference server, enabling efficient on-device generation.

**Stateful Orchestration** We utilize a graph-based orchestration framework to manage system execution, including state transitions and feedback loops. This enables persistent context tracking and iterative refinement of outputs.

**Vector Retrieval** A localized vector database is implemented using FAISS to support efficient semantic search. All embeddings are generated and stored locally to ensure data privacy and system autonomy.

### 5.3 Evaluation Dataset and Metrics

**Dataset Composition** We evaluate the framework on **150 organizational queries** spanning three categories:

- **Factual Queries (N=50):** Direct questions with verifiable answers in the knowledge base (e.g., "What is our company's internal compliance policy?")
- **Multi-Step Reasoning (N=75):** Complex queries requiring synthesis across multiple documents (e.g., "Compare last quarter's incident response times across departments")
- **Edge Cases (N=25):** Adversarial queries designed to trigger hallucinations (e.g., requests for non-existent information)

The knowledge base comprises **500 organizational documents** (policies, technical reports, meeting notes), totaling approximately 2.5M tokens. Documents were segmented into 300–500 token chunks following best practices for RAG systems [Liu et al., 2023].

**Evaluation Protocol** Each query was evaluated using DeepEval's RAG Triad metrics:

- **Faithfulness:** Detects contradictions between generated outputs and retrieved context (threshold  $\tau = 0.85$ )
- **Answer Relevancy:** Measures semantic alignment between responses and query intent
- **Contextual Recall:** Evaluates retrieval coverage of ground-truth information

A response was classified as hallucinated if the Faithfulness score was below 0.85 or if it introduced facts absent from the retrieved documents. All experiments were conducted using the Mistral-7B model with 4-bit GGUF quantization.

### 5.4 Reproducibility

To ensure reproducibility, we provide:

- **Code Repository:** Anonymous repository provided upon acceptance
- **Environment:** Docker container with exact dependency versions (Python 3.10, FAISS 1.7.4, LangGraph 0.1.2, DeepEval 0.21.x)
- **Model Weights:** Mistral-7B-Instruct-v0.2 GGUF (4-bit quantization)
- **Random Seeds:** Fixed at 42 for all stochastic operations
- **System Prompts:** Complete prompt templates provided in Appendix A

All experiments were conducted on a single machine with 16GB unified memory, demonstrating feasibility for edge deployment without specialized hardware.

## 6 Discussion

The experimental results demonstrate that system-level orchestration can significantly improve the reliability of small-scale foundation models without increasing parameter count.

**Accuracy vs. Model Scale** The transition from a high hallucination rate in the baseline configuration to near-zero hallucination in Phase 2 indicates that deterministic calibration ( $T = 0.1$ ) and contextual grounding have a substantial impact on factual consistency. This finding suggests that system-level design can be more effective than model scaling for improving reasoning reliability, particularly in resource-constrained environments.

**Latency vs. Reliability Trade-off** The introduction of stateful orchestration in Phase 3 results in an increase in average latency. While this introduces additional computational overhead, it provides a critical reliability mechanism by enabling iterative refinement and validation of outputs prior to final delivery.

**Systemic Efficiency and Robustness** The reduction in query rounds (NQ) observed in our results reflects efficiency gains similar to those reported in multi-agent collaboration frameworks [Zhang et al., 2025]. Advantage-based filtering serves as a more efficient grounding mechanism than standard physical verification, as it reduces reliance on repeated, unsuccessful environmental interactions. Furthermore, the framework’s ability to adapt during state transitions is supported by findings that advantage feedback enables rapid re-planning. When encountering unexpected disturbances, the advantage score functions as a reasoning anchor, allowing the agent to recalibrate its trajectory based on the current state’s potential rather than relying on outdated historical context [Zhang et al., 2025, Li et al., 2023].

**Deployment Feasibility** The system operates within a constrained memory budget of approximately 6.5GB on a 16GB unified memory system, demonstrating the feasibility of private, on-device AI systems without reliance on external cloud infrastructure [Zheng et al., 2025].

## 6.1 Failure Mode Analysis

Despite achieving near-zero hallucination rates, the system exhibits several limitations under specific conditions.

**Knowledge Base Gaps** When queries require information absent from the knowledge base, the system may generate responses based on parametric knowledge (8/150 test queries, 5.3%). For example, queries referring to policy changes introduced after the knowledge base snapshot may lead to unsupported or outdated responses.

**Ambiguous Query Intent** Vague queries lacking clear success criteria can trigger excessive self-correction loops ( $> 3$  iterations in 12/150 queries, 8.0%). In such cases, the system lacks explicit termination signals, making it difficult to determine when the reasoning process is complete.

**Multi-Hop Reasoning Limits** Queries requiring more than four reasoning steps exhibit degraded performance, with accuracy dropping from 98% to approximately 91.2% on complex test cases. This suggests limitations in maintaining coherent state across extended reasoning chains.

## 7 Conclusion

This work presented a modular framework for transforming baseline foundation models into grounded, stateful, and adaptive intelligent systems. By integrating deterministic calibration, contextual grounding, and graph-based orchestration, the proposed approach improves reasoning reliability without increasing model scale.

Experimental results demonstrate that the framework significantly reduces hallucination rates while enabling robust self-correction through stateful orchestration. Although this introduces additional latency, it provides a critical reliability mechanism for high-stakes and security-sensitive applications.

Overall, this work highlights the effectiveness of system-level design as an alternative to model scaling for achieving reliable and context-aware AI behavior. Future work will focus on optimizing verification loops to reduce latency and extending the framework to distributed multi-agent environments.

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